Asymptotic representation of the boundary-diffraction wave for a three-dimensional Gaussian beam incident upon a Kirchhoff half-screen

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Diffraction of a Gaussian beam with a circular spot size normally incident upon a Kirchhoff half-screen is investigated based on the boundary-diffraction-wave (BDW) theory. The evaluation of the boundary-diffraction wave by the steepest-descent method yields the uniform asymptotic representation of the total diffracted field consisting of the geometrical-optics and diffraction components. The use of complex rays to construct the diffraction component is also shown by applying the geometrical theory of diffraction (GTD) to the problem. Comparison of the representation of the diffraction component by the BDW theory with that by the GTD gives the diffraction coefficient for the Gaussian beam that ensures the continuity of the diffracted field at the shadow boundary.

INTRODUCTION

Ray-tracing techniques for studying propagation and diffraction of high-frequency fields have been examined by many authors. For homogeneous waves, such as plane, cylindrical, and spherical waves, Keller introduced a concept of a diffracted ray into geometrical optics to describe diffraction phenomena. His theory is known as the geometrical theory of diffraction (GTD).[1,2] To overcome the defect of the GTD, i.e., the field singularities at the shadow and reflection boundaries, the uniform GTD was developed by Kouyoumjian and Pathak.[3] These theories yield the asymptotic high-frequency solutions of diffraction problems.

In millimeter-wave and optical transmission systems, bounded beams are often used. Choudhary and Felsen[4,5] proposed the inhomogenous wave-tracking method, which takes place entirely in real space, and analyzed scattering of an evanescent plane wave and a Gaussian beam by a conducting circular cylinder. Keller and Streifer[6] showed that these inhomogenous waves are described in terms of complex rays. Wang and Deschamps[7] applied the complex ray-tracing technique to scattering of an evanescent plane wave by a conducting circular cylinder and compared their result with that deduced by the method of Choudhary and Felsen.[4]

It may also be necessary to study diffraction of inhomogeneous waves by obstacles with edges. Some papers have dealt with diffraction of a Gaussian beam by a circular aperture in a Kirchhoff screen[8–10] and a conducting half-screen[11] and have given the relationship between their results and ray optics. Those problems, however, are the two-dimensional problem in substance. Pearson *et al.*[12] discussed diffraction of a three-dimensional Gaussian beam by a Kirchhoff half-screen and showed good agreement between the scalar Kirchhoff–Fresnel diffraction theory and experiment. The relationship between their result and ray optics is, however, not made clear.

This paper treats diffraction of a three-dimensional Gaussian beam with a circular spot size normally incident upon a Kirchhoff half-screen. The analysis is based on the boundary-diffraction-wave (BDW) theory developed by Miyamoto and Wolf[13,14] and is restricted to the paraxial region because of the validity of the representation of the Gaussian beam. The total diffracted field, which consists of the geometrical-optics and diffraction components, is represented as a uniform asymptotic expression that ensures the continuity of the field at the shadow boundary defined by the Stokes line.

It is well known that the field of a Gaussian beam is paraxially equivalent to that of a spherical wave, with a source point having complex coordinates, which is described in terms of complex rays.[15,16] The diffraction component is also derived by analytic continuation of the half-screen in complex space and by applying the GTD to the bundle of complex rays describing the Gaussian beam. This expression is compared with the result from the BDW theory. The time factor $\exp(-i\omega t)$ is assumed.

1. BOUNDARY-DIFFRACTION WAVE

Let a Gaussian beam be normally incident upon a Kirchhoff screen in a half-plane $x > a, z = z_0$, as shown in Fig. 1. The field of the incident Gaussian beam as a bundle of complex rays is characterized by a complex beam parameter q as follows[17]:

$$U_i(x,y,z) = \frac{q(0)}{q(z)}\exp[ik\phi(x,y,z)], \tag{1}$$

where

$$\phi(x,y,z) = z + (x^2+y^2)/2q(z),$$

$$q(z) = z - ib/2,$$

$$b = kw_s^2, \qquad (k = 2\pi/\lambda), \tag{2}$$

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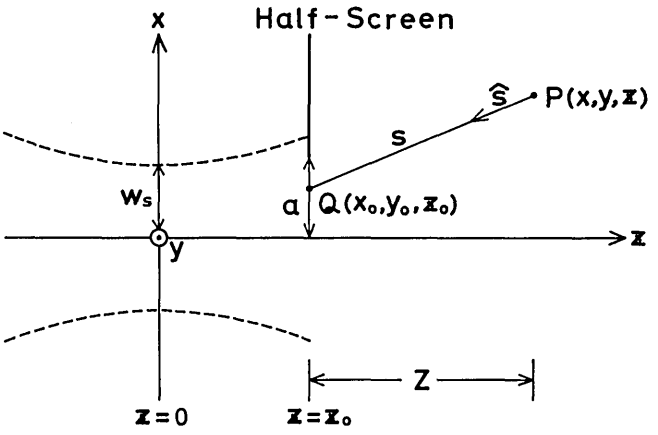


Fig. 1. Geometry of the problem.

and λ is the wavelength. The incident beam is assumed to propagate along the z direction and to have the smallest spot size w_s at $z = 0$.

Following the BDW theory,[13,14] we can represent the diffracted field $U_K(P)$ at an observation point P as

$$U_K(P) = \sum_j F_j(P) + U_B(P). \tag{3}$$

The first term in Eq. (3) contains the contributions from the singularities of a vector potential $\mathbf{W}(P, Q)$ on a plane of the aperture $x < a$, $z = z_0$. Within the paraxial approximation $x_0^2 + y_0^2 \ll |q(z_0)|^2$, $\lambda \ll w_s$, the vector potential $\mathbf{W}(P, Q)$ for the Gaussian beam is given by[8]

$$\mathbf{W}(P, Q) = U_i(Q)\frac{e^{iks}}{4\pi s}\frac{\hat{s}\times\nabla_Q\phi}{1+\hat{s}\cdot\nabla_Q\phi}, \tag{4}$$

where s is the distance between the observation point P(x, y, z) and a point Q(x_0, y_0, z_0) on the plane of the aperture and $\hat{s}$ is the unit vector in the direction from P to Q. The notation ∇_Q is used as the gradient operator taken with respect to the coordinates of Q, i.e.,

$$\nabla_Q\phi = \frac{x_0}{q(z_0)}\hat{x} + \frac{y_0}{q(z_0)}\hat{y} + \left[1 - \frac{x_0^2 + y_0^2}{2q^2(z_0)}\right]\hat{z}. \tag{5}$$

Because of the validity of the representation of the Gaussian beam and the BDW theory for a Kirchhoff screen, we should discuss the problem in the paraxial region: $|x_0 - x|^2 + |y_0 - y|^2 \ll |z_0 - z|^2$. Hence s and $\hat{s}$ are approximated as

$$s \simeq Z + [(x_0 - x)^2 + (y_0 - y)^2]/2Z \quad \text{(for phase term)},$$

$$s \simeq Z \quad \text{(for amplitude term)},$$

$$\hat{s} \simeq [(x_0 - x)\hat{x} + (y_0 - y)\hat{y} - Z\hat{z}]/Z, \tag{6}$$

where $Z = (z - z_0)$ is the distance between point P and its projection on the screen, and the vectors $\hat{x}$, $\hat{y}$, and $\hat{z}$ are the unit vectors in the x, y, and z directions, respectively.

The singularities of the vector potential $\mathbf{W}$ are determined from the denominator of the right-hand side of Eq. (4):

$$s(1 + \hat{s}\cdot\nabla_Q\phi) = \frac{1}{2Z}\left\{\left[x_0\frac{q(z)}{q(z_0)} - x\right]^2 + \left(y_0\frac{q(z)}{q(z_0)} - y\right)^2\right\} = 0. \tag{7}$$

From Eq. (7), two singular points $P_1(x_0^1, y_0^1, z_0)$ and $P_2(x_0^2, y_0^2, z_0)$ of the vector potential $\mathbf{W}$ on the plane of the aperture in real space are given by

$$x_0^1 = \frac{x - \gamma y}{1 + \gamma^2}, \qquad y_0^1 = \frac{\gamma x + y}{1 + \gamma^2}, \tag{8a}$$

$$x_0^2 = \frac{x + \gamma y}{1 + \gamma^2}, \qquad y_0^2 = \frac{-\gamma x + y}{1 + \gamma^2}, \tag{8b}$$

where γ is the ratio of the imaginary part to the real part of $q(z)/q(z_0)$.

Hence it is assumed that the y and z coordinates are fixed and that point P moves along the x direction. It can easily be shown that the contribution from each singularity P_1 and P_2 is equal to a half of the unperturbed incident beam $U_i(P)$ at P. Therefore the first term in Eq. (3) is represented as

$$\sum_{j=1}^{2} F_j(P) = [E(x_h^1 - x) + E(x_h^2 - x)]\, U_i(P)/2, \tag{9}$$

where $E(\tau)$ is the unit step function and x_h^1 and x_h^2 are the values of x when the singularities P_1 and P_2, respectively, are on the boundary of the half-screen, i.e.,

$$x_h^1 = a(1 + \gamma^2) + \gamma y, \tag{10a}$$

$$x_h^2 = a(1 + \gamma^2) - \gamma y. \tag{10b}$$

The second term $U_B(P)$ in Eq. (3) is the boundary-diffraction wave given by

$$U_B(P) = \int_\Gamma \mathbf{W}(P, Q)\cdot\hat{l}\,dl, \tag{11}$$

where $\hat{l}$ is the unit vector tangential to the boundary Γ of the half-screen and the contour integral should be taken counterclockwise when viewed from P. In derivation of Eq. (11), the contribution from an infinite semicircle centered at $(a, 0, z_0)$ below the boundary Γ is neglected because the field of the incident Gaussian beam vanishes rapidly away from the propagation axis. By substituting Eqs. (5) and (6) into Eq. (4), $U_B(P)$ is represented as

$$U_B(P) = \int_{-\infty}^{\infty} G(y_0)\exp[ikd(y_0)]dy_0, \tag{12}$$

where

$$d(y_0) = q(z_0) + \frac{a^2 + y^2}{2q(z_0)} + Z + \frac{(a - x)^2 + (y_0 - y)^2}{2Z},$$

$$G(y_0) = \frac{q(0)q(z_o)[x - a - aZ/q(z_0)]}{2\pi q^2(z)(y_0 - y_p^1)(y_0 - y_p^2)}e^{kb/2}, \tag{13}$$

and where y_p^1 and y_p^2 are poles of the integrand in Eq. (12) given by

$$y_p^1 = \frac{q(z_0)}{q(z)}y - ia\left[1 - \frac{q(z_0)}{aq(z)}x\right], \tag{14a}$$

$$y_p^2 = \frac{q(z_0)}{q(z)}y + ia\left[1 - \frac{q(z_0)}{aq(z)}x\right]. \tag{14b}$$

In deriving $G(y_o)$ in Eq. (13), the terms of $|y_0/q(z_0)|^2$ and $|a/q(z_0)|^2$ are neglected in comparison with unity because of the paraxial approximation. The length $d(y_0)$ is paraxially equal to the complex distance from the point $\overline{O}(0, 0, ib/2)$ to observation point P through point (a, y_0, z_0), i.e.,

$$d(y_0) \simeq [a^2 + y_0^2 + (z_0 - ib/2)^2]^{1/2} + [(a-x)^2 + (y_0 - y)^2 + (z_0 - z)^2]^{1/2}. \quad (15)$$

2. ASYMPTOTIC REPRESENTATION OF THE DIFFRACTED FIELD

It is difficult to obtain analytically the closed form of the boundary-diffraction wave given by Eq. (12). In high-frequency regions, however, the wave number k is so large that the intergral given by Eq. (12) can be evaluated approximately by the steepest-descent method.[18] The major contributions to $U_B(P)$ come from the poles $y_p{}^1$ and $y_p{}^2$ and the saddle point

$$y_s = [q(z_0)/q(z)]y. \quad (16)$$

Let x_s denote the value of x when two poles $y_p{}^1$ and $y_p{}^2$ are on the steepest-descent path (SDP), i.e., x_s is determined from $\mathrm{Re}[d(y_p{}^1)] = \mathrm{Re}[d(y_s)]$; then

$$x_s = a\left[\frac{4z^2 + b^2}{4z_0 z + b^2} + \frac{2b(z - z_0)}{4z_0 z + b^2}\left\{\frac{4z^2 + b^2}{4z_0^2 + b^2}\right\}^{1/2}\right]. \quad (17)$$

We define $x = x_s$ as the shadow boundary. Note that the poles $y_p{}^1$ and $y_p{}^2$ are real when $x = x_h{}^1$ and $x = x_h{}^2$, respectively. Since the coordinates $x_h{}^1$, $x_h{}^2$, and x_s depend on the location of observation point P, there are various cases in regard to the relative magnitudes of $x_h{}^1$, $x_h{}^2$, and x_s. The case in which $x_h{}^2 < x_s < x_h{}^1$ is shown in Fig. 2. Two poles $y_p{}^1$ and $y_p{}^2$ are points of symmetry with respect to saddle point y_s in the complex y_0 plane [see Eq. (14)]. As the x coordinate of observation point P increases, two poles move along the straight dashed line l_1 and l_2 in the direction of the arrow. The gradients of these lines depend on the z coordinate of point P.

In the deformation of the path of integration from $-\infty$ to ∞ into the SDP, some poles are encountered. The number of poles encountered depends on the location of observation point P. In the case in which $x_h{}^2 < x_s < x_h{}^1$, as shown in Fig. 2, it is found that no pole is encountered for region $x < x_h{}^2$, pole $y_p{}^2$ for $x_h{}^2 < x < x_s$, pole $y_p{}^1$ for $x_s < x < x_h{}^1$, and no pole for $x_h{}^1 < x$. Therefore the boundary-diffraction wave is represented as

$$U_B(P) = [E(x - x_h{}^1) + E(x - x_h{}^2) - 2E(x - x_s)]2\pi i\kappa \exp[ikd(y_p{}^1)] + \int_{SDP} G(y_0)\exp[ikd(y_0)]dy_0, \quad (18)$$

where κ is the residue of $G(y_0)$ at pole $y_p{}^1$,

$$\kappa = -\frac{iq(0)}{4\pi q(z)} e^{-kb/2}. \quad (19)$$

In derivation of Eq. (18), we took into account that the residue of $G(y_0)$ at pole $y_p{}^2$ is $-\kappa$ and that the complex distance $d(y_p{}^2)$ is equal to $d(y_p{}^1)$. It is easily shown that Eq. (18) holds for any relative magnitudes of $x_h{}^1$, $x_h{}^2$, and x_s.

Since the complex distance $d(y_p{}^1)$ is rewritten as

$$d(y_p{}^1) = \phi(x, y, z) - ib/2, \quad (20)$$

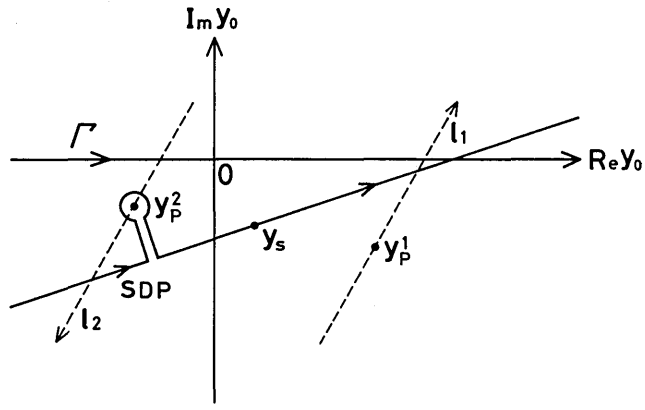


Fig. 2. Complex y_0 plane and contours of integration.

the contribution from the residue of $G(y_0)$ at pole $y_p{}^1$ is equal to one half of the unperturbed incident beam at point P, i.e.,

$$2\pi i\kappa \exp[ikd(y_p{}^1)] = U_i(P)/2. \quad (21)$$

As is shown in Appendix A, the integral along SDP in Eq. (18) is evaluated asymptotically as follows[18]:

$$\int_{SDP} G(y_0)\exp[ikd(y_0)]dy_0 \sim \mp 2\pi i\kappa \exp[ikd(y_p{}^1)]\mathrm{erfc}\left(\pm i\{ik[d(y_s) - d(y_p{}^1)]\}^{1/2}\right) + \sqrt{\pi/k}\exp[ikd(y_s)]\left(G(y_s)\left[\frac{i2q(z_0)Z}{q(z)}\right]^{1/2} + \frac{2\kappa}{\{i[d(y_s) - d(y_p{}^1)]\}^{1/2}}\right), \quad (x \lesssim x_s), \quad (22)$$

where $\mathrm{erfc}(\tau)$ is the complementary error function defined by

$$\mathrm{erfc}(\tau) = \frac{2}{\sqrt{\pi}}\int_\tau^\infty e^{-x^2}dx. \quad (23)$$

After some calculation of the right-hand side of Eq. (22), it is found that the second term cancels the third term, so that the only first term remains.

Use of Eqs. (9), (21), and (22) in Eq. (3) yields the uniform asymptotic representation of the diffracted field $U_K(P)$ as follows:

$$U_K(P) \sim U_G(P) + U_D(P), \quad (24)$$

where $U_G(P)$ and $U_D(P)$ are the geometrical-optics and diffraction components, respectively. These are given by

$$U_G(P) = E(x_s - x)\, U_i(P), \quad (25)$$

$$U_D(P) = D_u\left[\frac{q(z_0)}{q(z)Z}\right]^{1/2}\exp[iks(Q_d)]U_i(Q_d), \quad (26)$$

where $s(Q_d)$ is the complex distance from saddle point $Q_d(a, y_s, z_0)$ to observation point P,

$$s(Q_d) = Z + \frac{(x-a)^2 + (y - y_s)^2}{2Z}. \quad (27)$$

The first factor D_u in Eq. (26) is given by

$$D_u \mp \frac{1}{2}\operatorname{erfc}\left(\pm i\{ik[d(y_s)-d(y_p{}^1)]\}^{1/2}\right) \times \left[\frac{q(z_0)Z}{q(z)}\right]^{1/2} \exp\{ik[d(y_p{}^1)-d(y_s)]\}, \quad (x \lesseqgtr x_s). \tag{28}$$

By noting that $\operatorname{erfc}(-\tau) = 2 - \operatorname{erfc}(\tau)$, the diffracted field $U_K(P)$ given by Eq. (24) is found to be continuous at the shadow boundary $x = x_s$. Since the complementary error function is expressed asymptotically as

$$\mp\operatorname{erfc}(\pm\tau) \sim -\frac{e^{-\tau^2}}{\sqrt{\pi}\tau}, \qquad (|\tau| \gg 1), \tag{29}$$

the factor D_u becomes the following factor D asymptotically away from the shadow boundary:

$$D = \frac{e^{i\pi/4}\,2q(z_0)Z}{2\sqrt{2\pi k}[q(z)a - q(z_0)x]}. \tag{30}$$

3. INTERPRETATION OF THE DIFFRACTION COMPONENT IN TERMS OF THE GEOMETRICAL THEORY OF DIFFRACTION

In this section, we derive the representation of the diffraction component by applying the GTD to complex rays forming the Gaussian beam, and we compare it with that obtained from the BDW theory in Section 2.

Following the GTD, the field $\overline{U}_D(P)$ associated with the diffracted ray of a spherical wave by a straight edge is expressed as[1,2]

$$\overline{U}_D(P) = \overline{D}\left[\frac{r}{(r+\rho)\rho}\right] e^{ik\rho}\, U_i(\overline{Q}_d), \tag{31}$$

where r and ρ denote the distances from the center of the spherical wave and observation point P to the point of diffraction $\overline{Q}_d$, respectively. The edge-diffraction coefficient $\overline{D}$ for a Kirchhoff screen is given by[2]

$$\overline{D} = \frac{e^{i\pi/4}(\cos\theta - \cos\alpha)}{2\sqrt{2\pi k}\sin\beta(\sin\theta + \sin\alpha)} = -\frac{e^{i\pi/4}\tan[(\theta-\alpha)/2]}{2\sqrt{2\pi k}\sin\beta}. \tag{32}$$

The expression of Eq. (32) is valid only when observation point P is not close to the shadow boundary. In Eq. (32), β is the angle between the incident ray and the positive tangent to the edge. As is shown in Fig. 3, angles α and θ are the angles between the projections of the incident and diffracted rays into the plane normal to the edge and the negative z direction, respectively.

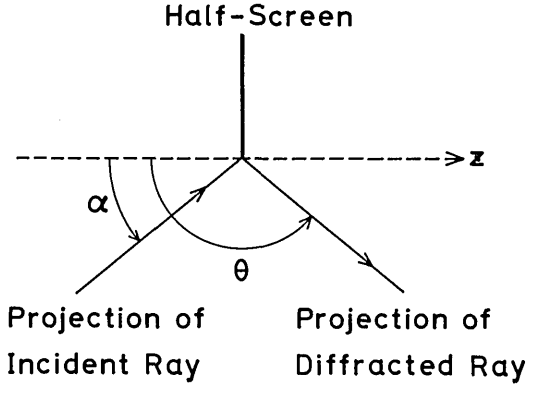


Fig. 3. Definition of the angles α and θ.

It has been shown that the field of a Gaussian beam is equivalent paraxially to that of a spherical wave with a center at a complex point $\overline{O}$.[15,16] The rectangular coordinates of point $\overline{O}$ are $(0, 0, ib/2)$ for the Gaussian beam represented by Eq. (1).

The complex diffraction point, which is on the complex extension of the edge $x = a, z = z_0$ of the half-screen, is determined according to the law of edge diffraction, i.e., the condition that the incident and diffracted rays must form an equal complex angle with the tangent at the edge. If y_d is the complex y coordinate of the point of diffraction $\overline{Q}_d$, the direction cosines in the y direction of the incident and diffracted rays are $y_d/q(z_0)$ and $(y - y_d)/Z$, respectively, within the paraxial approximation. Therefore the law of edge diffraction states that

$$\cos\beta = y_d/q(z_0) = (y - y_d)/Z. \tag{33}$$

We note that $Z = q(z) - q(z_0)$; Eq. (33) yields

$$y_d = [q(z_0)/q(z)]y. \tag{34}$$

This means that the diffraction point $\overline{Q}_d$ corresponds to the saddle point Q_d in Section 2.

Since the x and z components of the vectors $\bar{\mathbf{O}}\bar{\mathbf{Q}}_\mathbf{d}$ and $\overline{\mathbf{Q}}_\mathbf{d}\mathbf{P}$ are $[a, q(z_0)]$ and $[x - a, q(z) - q(z_0)]$, respectively, the tangent of the angle $(\theta - \alpha)/2$ is approximated as

$$\tan\left(\frac{\theta-\alpha}{2}\right) \simeq \frac{2q(z_0)[q(z)-q(z_0)]}{q(z)a - q(z_0)x}. \tag{35}$$

By using Eq. (35) and approximating the sine of β by unity because of the paraxial approximation, the diffraction coefficient $\overline{D}$ is found to coincide with the factor D in Eq. (30).

The complex distances r and ρ in Eq. (31) are approximated by

$$r \simeq q(z_0),$$

$$\rho \simeq q(z) - q(z_0) + \frac{(x-a)^2 + (y-y_d)^2}{2[q(z)-q(z_0)]} \quad \text{(for phase term)},$$

$$\rho \simeq q(z) - q(z_0) \quad \text{(for amplitude term)}. \tag{36}$$

Substitution of Eq. (36) into the second factor in Eq. (31) associated with the expansion ratio of the ray of the tube leads to

$$\left[\frac{r}{(r+\rho)\rho}\right]^{1/2} \simeq \left[\frac{q(z_0)}{q(z)[q(z)-q(z_0)]}\right]^{1/2}. \tag{37}$$

Moreover, the distance ρ (for the phase term) in Eq. (36) corresponds to $s(Q_d)$ in Eq. (27). Thus the diffraction component $\overline{U}_D$ given by Eq. (31) is rewritten as

$$\overline{U}_D \simeq D\left[\frac{q(z_0)}{q(z)Z}\right]^{1/2} \exp[iks(Q_d)]U_i(Q_d). \tag{38}$$

Comparing Eq. (26) with Eq. (38), the factor D_u is considered to be the diffraction coefficient for the Gaussian beam that ensures the continuity of the total diffracted field at the shadow boundary.

For diffraction of homogeneous waves, critical rays that graze the edge of the obstacle form the shadow boundary. There are also critical rays for diffraction of inhomogeneous waves described in terms of complex rays. In the following, we consider the critical rays for the case of the Gaussian beam represented by Eq. (1). The direct ray from the source point $\overline{O}$ $(0, 0, ib/2)$ to observation point P (x, y, z) intersects the aperture plane $(z = z_0)$ at point $Q_p(x_p, y_p, z_0)$, where

$$x_p = [q(z_0)/q(z)]x, \qquad y_p = [q(z_0)/q(z)]y. \tag{39}$$

From Eq. (7), it is found that this point Q_p corresponds to the singular point of the vector potential $\mathbf{W}$ in the complex $x_0 - y_0$ plane. Otis[8] showed that the contribution from the singularity Q_p is equal to the unperturbed incident beam $U_i(P)$ at observation point P. When point $Q_p(x_p, y_p, z_0)$ coincides with the diffraction point $\overline{Q}_d$ (a, y_d, z_0), the direct ray grazes the edge of the half-plane. The surface consisting of these critical rays is given by

$$x = [q(z)/q(z_0)]a. \tag{40}$$

However, these critical rays passing in complex space do not intersect real space. So we take $x = x_s$ given by Eq. (17) for the shadow boundary in real space according to the definition by Green *et al.*[11] From Eq. (20), the complex distance $d(y_p{}^1)$ is found to be equal to the distance of the direct ray from source point $\overline{O}$ to observation point P within the paraxial approximation. Therefore Green's definition of the shadow boundary means that the real parts of the complex distances along the direct ray and the ray through the diffraction point (that is, the increments of the real phases of two rays) are identical when point P is on the shadow boundary $x = x_s$.

Since the coefficient D_u becomes D away from the shadow boundary, as mentioned before, the diffraction component U_D based on the BDW theory agrees with that obtained from the GTD when the observation point is not close to the shadow boundary. This result implies that Keller's GTD is applicable to diffraction of complex rays as well as of real ones.

4. CONCLUSIONS

Diffraction of a three-dimensional Gaussian beam by a Kirchhoff half-screen has been discussed by using the boundary-diffraction-wave (BDW) theory. The uniform asymptotic representation for the diffracted field has been given by applying the steepest-descent method with a saddle point near a pole to the integral representation of the boundary-diffraction wave. Keller's geometrical theory of diffraction (GTD) was also been applied to the problem by treating the Gaussian beam as a bundle of complex rays issued from a complex point and by extending the half-screen into complex space. The diffraction coefficient for the Gaussian beam, which is valid at the shadow boundary, has been obtained by comparing two representations of the diffraction component based on the BDW theory and the GTD. The diffraction coefficient obtained here agrees with Keller's coefficient when the observation point is away from the shadow boundary. This suggests that Keller's GTD is an effective technique for diffraction of inhomogeneous waves as well as of homogeneous ones.

APPENDIX A. ASYMPTOTIC EVALUATION OF THE INTEGRAL ALONG THE STEEPEST-DESCENT PATH

The integration variable of the integral along SDP in Eq. (18) is changed from y_0 to ζ as follows:

$$\zeta^2 = i[d(y_s) - d(y_0)], \tag{A1}$$

such that the SDP transforms into the real axis of the complex ζ plane. The saddle point and poles $y_0 = y_s, y_p{}^1, y_p{}^2$ correspond to $\zeta = 0, \zeta_p, -\zeta_p$, respectively, where

$$\zeta_p = \{i[d(y_s) - d(y_p{}^1)]\}^{1/2}. \tag{A2}$$

The square root is defined so that ζ_p is positive real at $x = x_s$. On substitution of Eq. (A1) into Eq. (18), the integral along SDP becomes

$$\int_{\mathrm{SDP}} G(y_0)\exp[ikd(y_0)]\mathrm{d}y_0 = \exp[ikd(y_s)] \int_{-\infty}^{\infty} F(\zeta)e^{-k\zeta^2}\mathrm{d}\zeta, \tag{A3}$$

where

$$F(\zeta) = G(y_0)\frac{dy_0}{\mathrm{d}\zeta}. \tag{A4}$$

In order to take into account the effect of two poles $\zeta = \pm\zeta_p$ on the saddle point $\zeta = 0$, we set

$$F(\zeta) = \frac{\kappa}{\zeta - \zeta_p} - \frac{\kappa}{\zeta + \zeta_p} + H(\zeta), \tag{A5}$$

where $H(\zeta)$ is regular. Approximation of $H(\zeta)$ by $H(0)$ and use of the integral formula

$$\int_{-\infty}^{\infty} \frac{e^{-\Omega\xi^2}}{\xi^2 - \alpha^2}\mathrm{d}\xi = \mp i\frac{\pi}{\alpha}\,\mathrm{erfc}(\pm i\alpha\sqrt{\Omega})e^{-\Omega\alpha^2}, \qquad (\mathrm{Im}\,\alpha \lesseqgtr 0) \tag{A6}$$

yield the asymptotic representation of the integral along SDP, which is given by Eq. (22).

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